

$$1. \int \frac{4x^2 + 7x - 5}{x} dx = \int \left(4x + 7 - \frac{5}{x}\right) dx = \int 4x dx + \int 7 dx - \int \frac{5}{x} dx = 4 \int x dx + 7 \int 1 dx - 5 \int \frac{1}{x} dx = 4 \cdot \frac{x^2}{2} + 7 \cdot x - 5 \ln|x| + C = 2x^2 + 7x - 5 \ln|x| + C$$

$$2. \int \frac{3^x - 3^{2x} + 5^{x+2}}{4^x} dx = \int \left(\frac{3^x}{4^x} - \frac{3^{2x}}{4^x} + \frac{5^{x+2}}{4^x} \right) dx = \int \left(\frac{3}{4} \right)^x dx - \int \left(\frac{9}{4} \right)^x dx + 25 \int \left(\frac{5}{4} \right)^x dx = \frac{\left(\frac{3}{4} \right)^x}{\ln \frac{3}{4}} - \frac{\left(\frac{9}{4} \right)^x}{\ln \frac{9}{4}} + 25 \frac{\left(\frac{5}{4} \right)^x}{\ln \frac{5}{4}} + C$$

$$7. \int 22 \cdot (2x+3)^{10} dx = 22 \int (2x+3)^{10} dx = \left| \begin{array}{l} t = 2x+3 \\ dt = (2x+3)' dx \\ dt = 2 dx \\ dx = \frac{dt}{2} \end{array} \right| = 22 \int t^{10} \frac{dt}{2} = \frac{22}{2} \int t^{10} dt = 11 \cdot \frac{t^{11}}{11} + C = (2x+3)^{11} + C$$

$$9. \int \sqrt{3-2x} dx = \left| \begin{array}{l} t = 3-2x \\ dt = (3-2x)' dx \\ dt = -2 dx \\ dx = -\frac{dt}{2} \end{array} \right| = \int \sqrt{t} \left(-\frac{dt}{2} \right) = -\frac{1}{2} \int t^{\frac{1}{2}} dt = -\frac{1}{2} \cdot \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + C = -\frac{1}{2} \cdot \frac{2}{3} t \sqrt{t} + C = -\frac{1}{3} t \sqrt{t} + C = -\frac{1}{3} (3-2x) \sqrt{3-2x} + C$$

$$12. \int \frac{1}{\sin^2 \frac{2x+\pi}{3}} dx = \left| \begin{array}{l} t = \frac{2x+\pi}{3} \\ dt = \left(\frac{2x+\pi}{3} \right)' dx \\ dt = \frac{2}{3} dx \\ dx = \frac{3}{2} dt \end{array} \right| = \int \frac{1}{\sin^2 t} \cdot \frac{3}{2} dt = \frac{3}{2} \int \frac{dt}{\sin^2 t} = \frac{3}{2} (-\cot t) + C = -\frac{3}{2} \cot t = -\frac{3}{2} \cot \left(\frac{2x+\pi}{3} \right) + C$$